

CSE 416 Project Statement: MorningBrief.AI

TEAM MEMBERS

- Frontend & UI Designer : Dahyun Kwon
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- System Architecture & Admin : Minsoo Ahn

PROBLEM STATEMENT

In the modern information age, individuals are overwhelmed by a constant flood of news, trends, and data from countless sources. Rather than staying informed, many people experience "information fatigue", a state in which the sheer volume of content makes it difficult to distinguish what is relevant, reliable, or truthful. Despite this overload, the world continues to demand that people stay aware of current events, economic shifts, and daily lifestyle factors. Busy professionals and students, in particular, lack a simple, unified tool to receive the information that actually matters to them, without sorting through noise.

PROPOSED SOLUTION

MorningBrief.AI is a personalized AI-powered morning briefing web application that delivers a curated, minimalist daily digest tailored to each user's interests in under three minutes.

Upon first login via Google OAuth, users complete a short onboarding survey to define their preferences (e.g., finance, lifestyle, health). The AI engine then generates a personalized briefing covering relevant content such as today's weather and outfit recommendations, key economic indicators, trending news, food prices, and AI-suggested recipes based on uploaded refrigerator photos (media interaction).

Users access their dashboard each morning, at home or at the office, and immediately see a clean, widget-based summary of the world through the lens of what matters to them.

Unlike other AI assistants like OpenClaw, it does not include features that directly perform tasks to minimize security concerns. Furthermore, it's designed to be usable without requiring users to install a separate AI API, enhancing convenience and accessibility.

For the convenience of everyday users, it offers quick access and a user-friendly UI, lowering the barrier to entry for AI assistants that require complex features or device connectivity.

The platform supports two distinct user types:

- Standard Users : receive personalized AI briefings, upload media, and customize their dashboard.
- Admin Users : monitor system-wide usage statistics, manage content, and oversee platform health via a separate admin dashboard view

KEY FEATURES (A-Grade Requirements)

- Secure Login : Google OAuth 2.0 authentication
- Complex AI Logic : Survey-driven, multi-source AI briefing generation
- Media Interaction : Image upload for profile customization
- Responsive Design : Fully functional desktop browsers (possibly mobile in the future)
- Two User Views : Separate Standard User and Admin dashboards

DEVELOPMENT ROADMAP

Phase 1 — Foundation & Prototyping Core project setup, onboarding survey UI, basic dashboard layout.

Phase 2 — Core Features & AI Integration AI briefing engine (weather, news, economy widgets), Gemini/OpenAI API connection, Admin dashboard development.

Phase 3 — Google OAuth integration, Stabilization & A-Grade Feature Completion Media upload & interaction, security review, responsive optimization, beta release deployment.

Project management will follow the Scrum framework using **Jira**. Each sprint will consist of a defined set of tasks and todo lists.

TECH STACK (subject to change)

- Frontend : Next.js, HTML, CSS (Tailwind CSS)
- Backend : Express.js / FastAPI (Python), Google OAuth, Supabase (Database & Storage)
- Infra/Ops : Vercel (frontend deployment), Railway or Render (backend deployment)
- Project Mgmt: Jira (Scrum), GitHub (private repository)

** The above tech stack is tentative and subject to change based on team decisions and instructor feedback.*